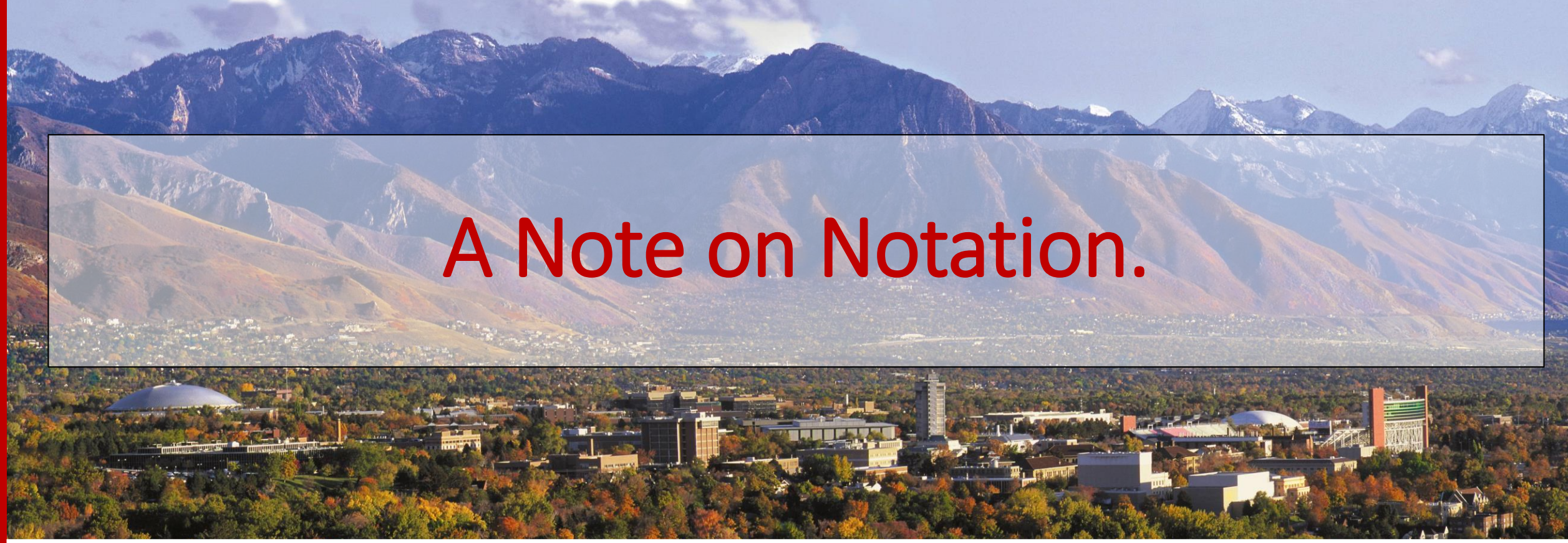

A Note on Notation.

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Formal Notation

- A big part of design and analysis is being able to translate our ***narrative*** hypotheses (e.g., there is a beneficial effect of exercise on resting blood pressure) into ***statistical*** hypotheses.

- In order to do this, we need to learn a bit about formal notation. Elaborating on our general framework of DATA = MODEL + ERROR, we can rewrite this as $y_i = \widehat{y_i} + \varepsilon_i$

  - $\widehat{y_i}$ is the estimated value for the ith observation in our data. This term could also be expanded to show exactly what the model is, not just the model's prediction: $\widehat{y_i} = \beta_0 + \beta_1(x_i)$

  - Thus, written more completely: $y_i = \beta_0 + \beta_1(x_i) + \varepsilon_i$

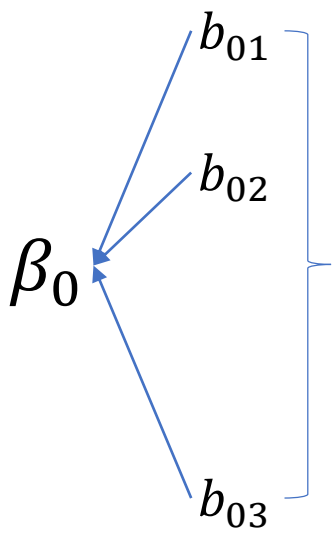
Formal Notation

- It is also important to distinguish between parameters (represented using Greek letters: $\beta, \sigma, \varepsilon$) and estimates of those parameters (represented using Roman letters: b, s, e).

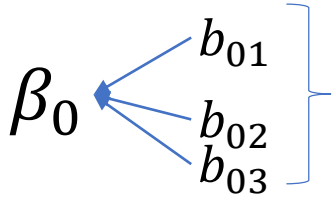
- The true value of the parameters are not knowable and thus need to be estimated from our sample.
  - If we had access to our entire population, the true values would be knowable, but this is almost never the case.

  - Usually, we are interested in applying the results of our study to large groups of people (e.g., all human adults; adults with a family history of heart disease; post-menopausal women)… and these groups are just too big to measure completely!

Parameter Estimates



Larger sampling variability in **Small** samples (e.g., n = 10)



Smaller sampling variability in **large** samples (e.g., n = 1,000)

- It is important to remember that these parameter estimates come from the data that we observe.
- They are informative, but if we had taken a different sample, the data would have also been different.
- All else being equal, the variability between samples depends on sample size (denoted by n).
  - Consider the effect of one outlier in a sample of 10 versus a sample of 10,000.

Sample Statistics

- As we will discuss, we want to use data from our sample to make inferences about likely values for parameters in the population.

- Often this means that we will need to calculate measures of *central tendency* and measures of *spread*:
  - Mean; $\bar{y} = \frac{\sum y_i}{n}$
  - Variance; $s^2 = \frac{\sum (y_i - \bar{y})^2}{(n-1)}$

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